

Eshat Jahan Arifa

B.Sc. in Biomedical Engineering, Chittagong University of Engineering & Technology (CUET)

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Academic Qualifications

Year	Degree & Institute	CGPA
2026	Bachelor of Science in Biomedical Engineering Chittagong University of Engineering & Technology (CUET), Bangladesh	3.62/4.00

Research Interests

Biomedical Signal Processing, Biosensors and Biomedical Sensing, Electrochemical Sensing, Wearable Devices, Biomedical Instrumentation, Medical Image Analysis, AI-Powered Predictive Healthcare Systems, Neural Engineering and Brain-Computer Interfaces, Bioinformatics, Biomaterials and Drug Delivery Systems, Nanotechnology and Nanomedicine.

Undergraduate Thesis

Development and Characterization of Natural Polymer-Based *Calendula officinalis* Loaded Hydrogel for Wound Dressing Application Nov 2025 – Jun 2026

Supervisor: Ayesha Binth Humayun

- Developed and characterized gelatin–sodium alginate hydrogels incorporated with *Calendula officinalis* extract for potential wound-dressing application.
- The composition shows better result then the existing samples.

Research Publications

- Md. Taisirul Muktadi, Tanzeem Tahmeed Reza, Mohammed Tawshif Hossain, **Eshat Jahan Arifa**, and Md. Fazlul Karim Khondakar, “Spectrogram-driven Emotion Detection from Electroencephalogram,” in *2025 4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, CUET, Chattogram, Bangladesh, IEEE, 2025. DOI: [10.1109/ECCE64574.2025.11013815](https://doi.org/10.1109/ECCE64574.2025.11013815). (Published)
- Tanzeem Tahmeed Reza, Sharafat Hossain, and **Eshat Jahan Arifa**, “Novel Task-Specific Features Extraction and Comparative Evaluation for Parkinson’s Detection from Digital Pen Signal Data,” in *2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, IEEE, 2025. DOI: [10.1109/BECITHCON69222.2025.11504012](https://doi.org/10.1109/BECITHCON69222.2025.11504012). (Published)

Research Experience

Research Assistant, Electrochemical Lab, CUET

Jul 2026 – Present

Supervisors: Dr. Sumit Majumder and Dr. Arup Kumer Roy

- Conducting electrochemical measurements and analyzing responses for hydrocortisone detection (work in progress).

Internship Experience

Intern, Tradevision LTD

Apr 2025 – May 2025

- Gained practical exposure to the operation and working principles of CT scanners, CT simulators, ultrasonography systems, dialysis machines, and ophthalmology equipment.
- Observed the operation of ICU facilities, cyclotron systems, medical gas pipeline systems, and hospital waste management practices.
- Developed practical understanding of biomedical equipment, hospital infrastructure, and healthcare

technology management.

Academic Projects

Consumer Choice Prediction Using EEG Signals 2025

- Developed an EEG-based machine learning framework to classify consumer decisions (Buy vs. No-Buy) using neural activity patterns from the NeuMa dataset.

Skin Lesion Boundary Detection Using Classical Image Processing 2024

- Developed a medical image segmentation pipeline to detect skin lesion boundaries from dermoscopic images using the PH2 dataset.

Movement Pattern Recognition and Phase Detection Using Wearable Sensors 2024

- Built a wearable motion classification system to identify standing, walking, and falling phases.

Digital Stethoscope for Real-Time Auscultation Monitoring 2024

- Designed a real-time biomedical monitoring architecture for capturing and wirelessly transmitting heart and lung sound data.

Smart Moisture Detection System for Bedridden Patients 2023

- Developed a low-cost embedded healthcare solution for detecting moisture conditions in bedridden patients.

Technical Skills

Programming & Libraries:	Python, C/C++, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib
Biomedical Computing & Laboratory	MATLAB, Simulink, EEGLAB, AutoDock Vina, PyMOL, Signal Processing, Medical Image Analysis, Machine Learning, Cyclic Voltammetry, Potentiostat Operation, Hydrogel Fabrication, In-vitro testing.
Documentation:	LaTeX / Overleaf, Microsoft Word, Microsoft Excel, Microsoft PowerPoint

Achievements & Certifications

- **Galactic Problem Solver**, NASA International Space Apps Challenge (2025)
- **Galactic Problem Solver**, NASA International Space Apps Challenge (2024)
- **Champion**, BioOlympiad – Department of Biomedical Engineering, CUET (2025)
- **Champion**, Idea Contest – Department of Biomedical Engineering, CUET (2025)
- **Champion**, Idea Contest – Department of Biomedical Engineering, CUET (2022)
- **Finalist**, Face the Case Competition – IEEE CUET Student Branch (2025)
- **Finalist**, Face the Case Competition – IEEE CUET Student Branch (2023)
- **Participant**, 3-Minute Thesis (3MT) Competition, SciBlitz 2.0 – IEEE CUET Student Branch (2026)

Extracurricular & Volunteer Activities

- **Joint General Secretary**, IEEE EMBS CUET Student Chapter (August 2025 – August 2026)
- **Assistant General Secretary**, BMES CUET (August 2025 – August 2026)

Language Proficiency

Medium of Instruction:	English
IELTS:	Scheduled to appear in October 2026